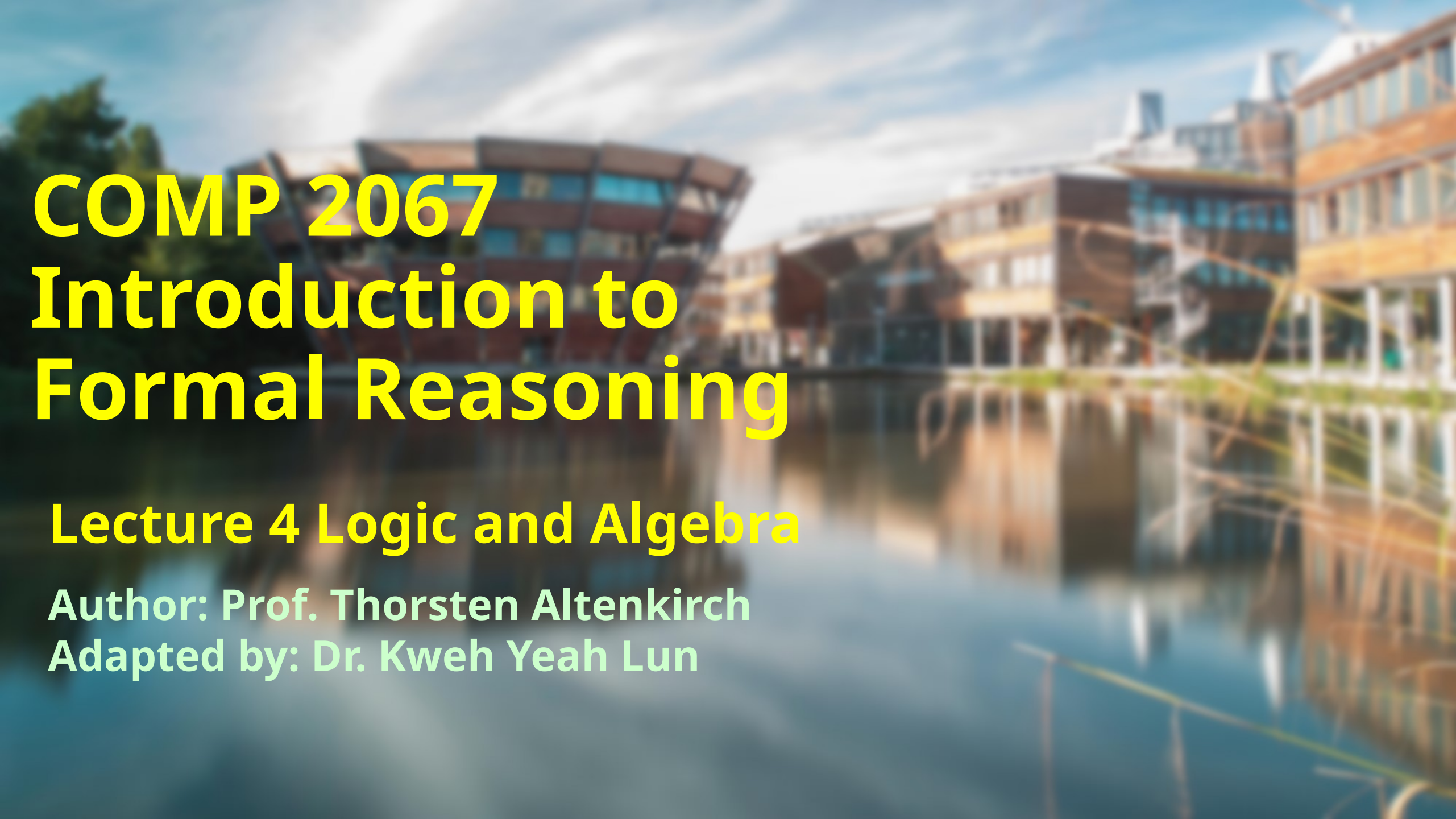




COMP 2067

Introduction to

Formal Reasoning

Lecture 4 Logic and Algebra

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Learning Outcome

At the end of this lecture, you will be able to

- explain the relationship between logic and algebra
- the concept of true, false and negation and how does it apply in Lean



Logic and Algebra

- In algebra: **distributive law** $x(y + z) = xy + xz$
- Similar law holds in propositional logic.

- **Exercise 1**

```
example : P ∧ (Q ∨ R) ↔ (P ∧ Q) ∨ (P ∧ R) :=  
begin  
  sorry  
end
```

[Try it!](#)



- In Lean you can say **sorry** and omit the proof.
- So, if you cannot complete your Lean exercise, you just type **sorry** and it is OK in Lean.
- However, I will say **sorry** too for your coursework marks. ◀◀



- Next, the **counterpart to implication is exponentiation** but have to **read it backwards**, that is $P \rightarrow Q$ becomes $Q^{\wedge}P$.
- Then the translation of the law $x^{yz} = (x^y)^z$ corresponds to the currying equivalence $P \wedge Q \rightarrow R \leftrightarrow P \rightarrow Q \rightarrow R$.
- There is another law of exponentiation
 $x^{y+z} = x^y x^z$
- Its translation is also a law of logic.



- Exercise 2:

```
theorem case_thm : P ∨ Q → R ↔ (P → R) ∧ (Q → R) :=  
begin  
  sorry  
end
```

[Try it!](#)

- Note that the right to left has been shown in the previous example [case_lem](#).



True, false and negation

- There are **two logical constants**: **true and false**.
- **Example**:
 1. It always shiny in the morning at KL. (True)
 2. Kweh can fly. (False)
- In logic and proof, **true is like an empty conjunction** and **false is an empty disjunction**.



- **True is like empty conjunction**
 - combining no statements with " $\wedge$ " doesn't affect the overall truth value, which remains simply "True".
 - **there is no statement to make it "false".**
 - **Analogy:** Imagine: "I will give you a grade A *if...*" and then say nothing else. I haven't specified any conditions. This is an **empty conjunction**. Since there are **no conditions to be false**, the **statement** (the promise) could be **considered "true"** in a very weak, almost meaningless sense.



- **False is like empty disjunction**

- there are no propositions to consider, so there are no conditions to be met for the disjunction to be true.
- **Analogy:** Scenario: A cafe offers a special: **Americano or Latte or Mocha**. If you **choose any of these**, you **get the special**. If the cafe says **the special is available if you choose...**, and then **lists nothing**, it means you **can't choose anything**. You're **not getting the special**. The statement "You get the special" is therefore **"False."**



- Example:

```
example : true :=  
begin  
  constructor,  
end
```

Try it!



- Symmetrically there is no way to prove false because we neither have left nor right.
- Doing cases on false as an assumption makes the current goal go away as by magic and leaves no goals to be proven.
- "cases on false" refers to the principle that from a contradiction (i.e., a proof of False), you can derive anything.
- This is sometimes called "ex falso quodlibet" (from falsehood, anything follows).



- If your goal is to prove something, and you find yourself in a situation where you have a case based on "false," it means you've essentially reached a dead end.

• **Example** `theorem efq : false → P :=`
 `begin`
 | `assume f,`
 | `cases f,`
 `end`

Try it!

- efq is short for the latin phrase *Ex falso quodlibet*.
- In logic, the reasoning **If Kweh can fly then I can passed COMP2048** is **valid**.



- $\neg P$ is defined as $P \rightarrow \text{false}$ which means that P is impossible.
- As an example, prove the law of contradiction: it cannot be that both P and $\neg P$ hold.
- Example

```
theorem contr:  $\neg (P \wedge \neg P) :=$   
begin  
  assume pnp,  
  cases pnp with p np,  
  apply np,  
  exact p,  
end
```

Try it!



Summary of tactics

	How to prove ?	How to use?
$\rightarrow$	assume h	apply h
$\wedge$	constructor	cases h with p q
$\vee$	left right	cases h with p q
true	constructor	
false		cases h



- Additionally, the tactic **assumption** checks whether any assumption matches the current goal.
- Hence, the first proof can be rewritten as:

```
theorem I : P → P :=  
begin  
  assume h,  
  assumption,  
end
```

[Try it!](#)



- Give yourself a big clap.
- You have survive until here. ◀◀
- Any question?



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Next lecture...

- Classical logic
- Intuitionistic logic



“I know what it's like to lose. To feel so desperately that you're right, yet to fail nonetheless.”

– Thanos. (2018). *Avengers: Infinity War*. Walt Disney Studio Motion Pictures.

Hope you all enjoy this lecture!